DOST Form 5

PROJECT WORKPLAN

(1) Program Title: n/a
(2) Project Title: Promoting Coastal Resource Conservation Activities by Local Communities through Fostering Social Capital: Approach Using Public Good Experiment
(3) Project Duration (number of months): 24 **(4) Project Start Date:** September 1, 2025 **(5) Project End Date:** August 30, 2027

(6) OBJECTIVES *(with % weight)*	(7) TARGET ACTIVITIES	(8) TARGET ACCOMPLISHMENT *(with % weight and objectively verifiable indicators)*	(9) YEARLY AND QUARTERLY TARGETS																	
			Y1 *(56.9%)* *(with % weight per year and quarter)*						Y2 *(43.1%)* *(with % weight per year and quarter)*						Y3 *(with % weight per year and quarter)*					
			Q1	Q2	Q3	Q4	Total	Weighted Yearly total	(d1) Q1	(d2) Q2	(d3) Q3	(d4) Q4	(d5) Total	Weighted Yearly total	(e1) Q1	(e2) Q2	(e 3) Q3	(e4) Q4	(e5) Total	Weighted Yearly total
1. To measure, assess, and compare the cooperative behavior among Indigenous and local fishers through the public goods experiment in the context of sustainable marine resource management *(26%)*	1. Design and administer the Public Good Experiment (PGE) to capture the cooperative behavior (CB) of the target fisher participants;	DeA well-structured and culturally sensitive public good experiment design; Recruitment of indigenous and local fisher participants following an appropriate sampling framework for each study site; Preparation and validation of economics experiment instruments, materials, and logistic arrangements for the PGE; Administration of the economics experiments to the indigenous and local fisher participants. *(10%)*	20 %	10 %	70 %		100 %	10.0 %												
	2. Conduct preliminary site visits, key informant interviews, and secondary data collection;	Completion of preliminary site visits, key informant interviews with stakeholders, and secondary data collection to gather contextual and logistic information, and other documentation;. *(4%)*	30 %	40 %	30 %		100 %	4.0 %												
	3. Forge MOU/MOA with partner institutions or stakeholder groups;	Draft of MOU/MOA documents; Stakeholder engagement and consultation meetings, documentation	80 %	20 %			100 %	4.0 %												

		of stakeholders' inputs and agreements to finalize the MOU/MOA; MOU/MOA signing with target stakeholder groups or partner institutions with the presence of representatives LGUs, indigenous groups and community leaders; Official documentation of signed agreements; *(4%)*																		
	4. Perform exploratory data analysis (EDA) and interpretation;	Dataset (CB dataset) prepared with clear documentation of variable definitions; Completion of EDA report summarizing key findings supported by descriptive statistics, visualizations, and interpretations; Initial insights and preliminary interpretation on cooperative behavior; *(4%)*				30%	30 %	1.2 %	50 %	20 %			70 %	2.8 %						
	5. Perform PGE-specific data analysis and interpretation.	Group-level and round-level analysis and visualization of PGE data, interpretation, documentation and reporting; *(4%)*				50%	50 %	2.0 %	50 %				50 %	2.0 %						
2. To analyze the effect of social capital (levels of trust and reciprocity) on cooperative behavior among Indigenous and local fishers in marine resource management *(12%)*	1. Design and administer the social capital surveys;	Development of culturally appropriate and validated social capital survey instrument; Selection and recruitment of indigenous and local fisher respondents following an appropriate sampling framework for each study site;	20 %	10 %	70 %		100 %	5.0 %												

		Administration of social capital surveys prior the economics experiments; *(5%)*																		
	2. Data integration and perform exploratory data analysis (EDA) and interpretation;	Dataset (SC dataset) prepared with clear documentation of variable definitions; Integration of SC data set with CB dataset, and completion of EDA report summarizing key results supported by descriptive statistics, visualizations and interpretations; *(2%)*				70%	70 %	1.4 %	20 %	10 %			30 %	0.6 %						
	3. Perform econometric and/or regression analysis;	A detailed econometric analysis report; Econometric models relating social capital and cooperative behavior; Documentation, comparative analysis, interpretation, and reporting of key model outputs; *(3%)*				70 %	70 %	2.1 %	20 %	10 %			30 %	0.9 %						
	4. Administer post-experiment social capital surveys and analysis	Completion of post-experiment surveys; Comparative data analysis and interpretation of pre and post experiment social capital survey data. Documentation, comparative analysis, interpretation, and reporting; *(2%)*		10 %	60 %	15 %	85 %	1.7 %	10 %	5 %			15 %	0.3 %						
3. To assess how cultural values (beliefs about communal responsibility,	1. Administer qualitative interviews on cultural values;	Completion of qualitative interviews identifying cultural beliefs, practices and perceptions;	40 %	20 %	40 %		100 %	2.0 %												

respect for nature, attitude towards marine resource sharing and management, etc.) of Indigenous and local fishers influence their social capital and cooperative behavior in marine resource management; ***(12%)***		***(2%)***																		
	2. Design and administer the cultural values surveys;	Development of cultural values survey instrument; Administration of cultural values surveys prior the economics experiments; ***(6%)***	20 %	10 %	70 %		100 %	6.0 %												
	3. Data integration and perform exploratory data analysis (EDA), qualitative data analysis, correlational analysis and interpretation;	Dataset (CV dataset) prepared with clear documentation of variable definitions; Merging CV data with CB and SC datasets, and completion of EDA report of CV data summarizing key results supported by descriptive statistics, visualizations and interpretations; Collection of context-specific qualitative data complementing the CV dataset, thematic analysis, and interpretation; Perform simple and/or canonical correlation analysis between cultural values, social capital, and cooperative behavior; ***(4%)***				60 %	60 %	2.4 %	25 %	10 %	5 %		40 %	1.6 %						
4. To examine the influence of socio-economic factors (age, income, educational level, fishing experience, etc.) on social capital and cooperative behavior in marine resource management; ***(10%)***	1. Design and administer socioeconomic attributes survey;	Development social socioeconomic attributes survey instrument; Administration of social capital surveys prior the economics experiments; ***(5%)***	40 %	20 %	40 %		100 %	5.0 %												
	2. Data integration and perform exploratory data analysis (EDA), segmentation analysis, and interpretation;	Dataset (SE dataset) prepared with clear documentation of variable definitions; Merging SE data with CB, SC and CV datasets,				70 %	70 %	1.4 %	20 %	10 %			30 %	0.6 %						

		and completion of EDA report of CV data summarizing key results supported by descriptive statistics, visualizations and interpretations; ***(2%)***																		
	3. Perform econometric analysis.	A detailed econometric analysis report;. Econometric models relating socioeconomic variables with social capital and cooperative behavior. ***(3%)***				70 %	70 %	2.1 %	20 %	10 %			30 %	0.9 %						
5. To provide recommendations for promoting sustainable marine resource management, especially within marine protected areas (MPAs), by enhancing social capital and fostering cooperative behavior, considering the role of incentives and the influence of cultural values. ***(40%)***	1. Synthesize findings;	Synthesize all research findings into a cohesive summary, highlighting key factors for promoting sustainable resource management; ***(2%)***								60 %	40 %		100 %	2.0 %						
	2. Focus group discussions;	Conduct focus group discussions, ***(2%)***							70 %	30 %			100 %	2.0 %						
	3. Benchmarking and consultations;	Conduct local and foreign benchmarking visits and consultations with experts; ***(3%)***			100 %		100 %	3.0 %												
	4. Develop policy recommendations;	Draft policy recommendations and best practices for enhancing social capital and cooperative behavior, with a focus on MPAs; ***(6%)***				10%	10 %	0.6 %	30 %	30 %	30 %		90 %	5.4 %						
	5. Community workshop;	Conduct community workshops to present findings, gather feedback, and refine recommendations based on local input; ***(5%)***								50 %	50 %		100 %	5.0 %						

	6. Create an action plan;	Develop actionable plans for implementing the recommendations; *(6%)*				10%	10 %	0.6 %	10 %	30 %	50 %		90 %	5.4 %						
	7. Develop manual;	Develop a comprehensive manual for conducting PGE in fishing communities for managing common-pool fishery resources; *(8%)*			5 %	5 %	10 %	0.8%	10 %	10 %	30 %	40 %	90 %	7.2 %						
	8. Dissemination	Publish and disseminate findings. *(8%)*				20 %	20 %	1.6 %	30 %			50 %	80 %	6.4 %						

(10) EXPECTED OUTPUTS (6Ps)	YEARLY AND QUARTERLY TARGETS															Grand Total
	Y1					Y2					Y3					
	Q1	Q2	Q3	Q4	Total	Q1	Q2	Q3	Q4	Total	Q1	Q2	Q3	Q4	Total	
Publications																
2 publishable article targeted for submission to high-impact, peer-reviewed journal					-			1	1	**2**						**2**
Patents/IP					-					-						
Products																
1 comprehensive manual for conducting PGE in fishing communities for managing common pool fishery resources					-				1	**1**						**1**
People Services																
90 extension beneficiaries to undergo training/ workshops, with a focus on fostering social capital, promoting cultural values, and developing cooperative behavior to enhance sustainable marine resource management practices)					-				90	**90**						**90**
Places and Partnerships																
6 MOUs/MOAs forged with LGUs and barangays to ensure strong partnership, active engagement and collaboration	6				**6**					-						**6**
Policy																
4 evidence-based policy recommendations, one tailored for each municipality involved, focusing on sustainable management of shared fishery resources					-				4	**4**						**4**

(11) Potential Impacts:	
Social Impact	**Economic Impact**
The study will elucidate the role of social capital and cultural values in fostering cooperative behavior within MPAs. By focusing on indigenous and local fishers, the research will enhance community engagement in resource management, reinforcing trust and reciprocity essential for effective collaboration. This increased participation is expected to strengthen social cohesion and collective action, leading to more resilient and self-sustaining communities. Furthermore, integrating traditional ecological knowledge into conservation strategies will preserve cultural heritage while aligning with modern management practices, thereby reinforcing community identity and resilience.	This study has the potential to improve the sustainability and productivity of marine resources, which are crucial to the livelihoods of coastal communities. By fostering effective MPA management through enhanced cooperation, the study aims to stabilize fish stocks and ensure long-term resource availability, directly benefiting local and indigenous fishers. Additionally, improved ecosystem health could boost ecotourism and related economic activities, diversifying income sources for these communities. The insights gained will inform policy decisions, leading to more efficient resource management and enhanced economic stability in coastal regions.

(12) Risks Management Plan

Objective	Risks	Risk Rating						(13) Mitigation/Action Plan (Use separate sheet if necessary)
		Impacts on Timeline		Impact on Budget		Probability		
		Rating	Reason	Rating	Reason	Rating	Reason	
Objective 1: Measuring cooperative behavior	Participant reluctance, cultural misunderstandings, data collection challenges.	Medium	Delays may arise due to difficulties in participant engagement and logistical challenges.	Low	Extra costs might be incurred for additional community engagement, but these are not substantial.	Medium	Participants' initial hesitation is likely but can be mitigated with proper planning.	Build trust with communities, adapt the experiment to local contexts, and prepare logistics in advance.
Objective 2: Effect of social capital on cooperative behavior	Inaccurate measurement of trust and reciprocity, response bias.	Low	Additional testing of survey tools might extend timelines slightly.	Low	Pre-testing does not involve major expenses.	Medium	Challenges in measuring trust and reciprocity are common.	Use mixed methods - surveys and interviews, pre-test tools.
Objective 3: Influence of cultural values	Cultural insensitivity, overgeneralization, misinterpretation of data.	Medium	Delays may occur while refining tools and analyzing data within the cultural context.	Medium	Additional costs for cultural training and engaging local experts are likely to arise.	Medium	Risks are manageable but require proactive actions to mitigate cultural misinterpretations.	Provide cultural training, involve local experts, and diversify sample groups.
Objective 4: Socio-economic factors impact	Inaccurate socio-economic data, multicollinearity.	Medium	Robust data validation may require additional time to ensure accuracy.	Low	Implementing statistical techniques to address multicollinearity will incur minimal costs.	Medium	Socio-economic data challenges are common but can be addressed with careful planning.	Ensure anonymity, use robust statistical techniques, and conduct context-specific analyses.

Objective 5: Recommendation for sustainable management	Policy implementation challenges, lack of impact, overemphasis on incentives.	Medium	Delays may occur due to prolonged stakeholder engagement and policy formulation.	Medium	Communication and engagement efforts may increase costs.	Medium	Translating research findings into actionable policies is a common challenge.	Engage stakeholders early, develop clear communication strategies, and provide a holistic approach.

Legend:
Impact on Time Rating: Low (minimal delay in days); Medium (moderate delay in weeks or months); High (significant delays in year)
Impact on Budget Rating: Low (<5% increase in budget); Medium (5-10% increase in budget); High (>15% increase in budget)
Probability Rating: Low (<30% chance of occurrence); Medium (30-70% of occurrence); High (>70% chance of occurrence)

Prepared by: **JOELA MIZCHELLE D. DACILLO, PhD**
Project Leader

Noted by: **ARNEL B. ZARCEDO, EDEL**
Head of Implementing Agency